

OPTIMA-FLEET: Optimization and Machine-learning Analytics for Freight Logistics Efficiency Leveraging Emerging Technologies

Overview: Tennessee stands as one of the national leaders in freight and logistics, hosting major transportation hubs and extensive highway corridors such as I-40, I-24, I-75, I-65, I-55, I-240 and others. The state's strategic location and infrastructure make it a vital logistics center for renowned firms like FedEx, UPS, J.B. Hunt, and Old Dominion, facilitating freight distribution across the Southeast. However, the state's freight network faces significant challenges, including congestion, inefficient routing, limited real-time operational insights, and first-last mile inefficiencies. These challenges result in increased delays, higher operational costs, elevated crash rates involving freight vehicles, and excessive fuel consumption leading to environmental concerns. Moreover, the lack of integrated and real-time decision-making capabilities exacerbates the inefficiencies in freight coordination, particularly impacting the seamless connectivity between intermodal hubs, warehouses, and final delivery points. Also, truck parking on interstate ramps has emerged as a growing problem in Tennessee, posing significant safety and maintenance challenges. The lack of adequate off-site parking facilities during night and peak hours forces many truck drivers to park on interstate ramps, increasing the risk of accidents and complicating traffic operations. This forces truck drivers to either park illegally on highway shoulders and ramps or continue driving in violation of federal Hours-of-Service (HOS) regulations, both of which pose serious safety risks. In response to these challenges, by leveraging partnership with TDOT Freight Planning Division, this project proposes the development of a predictive analytics system that leverages real-time and historical freight data, AI-based forecasting, and economic impact analysis to optimize freight movement across Tennessee.

Objectives and Goals: The primary objective of this project is to establish an AI-driven freight analytics platform that enhances the efficiency, safety, and environmental sustainability of freight operations in Tennessee. The project seeks to optimize freight routing by developing predictive models capable of reducing congestion and improving routing decisions. Enhancing road safety is also a key goal, with the use of AI-driven crash prediction models to proactively mitigate risks on key freight corridors. By integrating eco-routing algorithms, the project aims to reduce fuel consumption and greenhouse gas emissions, contributing to Tennessee's sustainability goals. Additionally, the project will foster public-private collaborations with stakeholders such as TDOT Freight Planning and leading logistics firms, while also leveraging AI to bolster supply chain resilience through adaptive freight scheduling and real-time adjustments.

Scope: The scope of this project encompasses a comprehensive analysis of freight traffic operations, predictive analytics, safety enhancements, and the economic impacts of optimized freight logistics. Spanning a period of two years, the project will focus on leveraging AI and machine learning to address bottlenecks in freight movement, improve real-time operational decision-making, and streamline first-last mile logistics. The project will also address parking challenges through developing data-driven solutions that can optimize parking availability, improve driver compliance and safety, and ensure uninterrupted freight movement. By targeting key freight corridors and integrating data-driven solutions, the project aims to establish a scalable model for AI-driven freight optimization that can be replicated across Tennessee.

Task 1: Comprehensive Literature Review

A comprehensive literature search will be undertaken to uncover both ongoing or previous published and unpublished reports and papers on Freight transportation challenges, and any other relevant materials on this subject. The review will also help to determine information and practices conducted

by other states, which are also relevant to this study. The review will also cover methodologies that are used for analysis, evaluation and solving freight issues as well as tools, algorithms, software and models for mitigating freight issues. The resources will include library holdings, databases, and gateway services. Through these libraries, external database services such as TRB, ASCE, ITE, NCHRP, TRIS, Elsevier Science, and others will be accessed.

Task 2: Freight Corridors and Connectors Inventory

Data collection will begin with a review of available databases and studies from TDOT and stakeholders. A field review of the known freight corridors, segments and connectors will be conducted. The field review will be aimed at seeking input regarding the study's focus, specific areas of concern related to freight corridors. The field review may include a windshield survey of the facilities looking for obvious signs of deficiencies like tire marks on curbs, indications of queue storage lengths being exceeded, and delay at intersections.

Task 3: Identification of Freight Deficiencies

Analysis will be conducted to identify deficiencies and issues along selected freight corridors and connectors based on the inventories gathered data. Analysis will include evaluation of factors such as adequate capacity and access to hubs, turning radii, queuing storage, pavement quality, and freight access limitations.

Task 4: Data Collection and Gathering

This task will involve comprehensive data collection to support the development of predictive models for freight logistics optimization. The data gathered will include traffic flow, intersection turning volumes, historical traffic data, freight movement data, freight intermodal connectors data, safety data, and GIS data. Additionally, information on truck parking availability and patterns will be collected alongside questionnaire surveys from truck drivers and fleet managers to capture insights on parking challenges and operational needs. This diverse dataset will form a robust foundation for training AI-driven models aimed at enhancing safety, efficiency, and real-time decision-making in Tennessee's freight logistics network.

Task 5: Engaging Freight Partners and Stakeholders and Survey

To better understand the context of corridors and clarify specific freight issues and concerns, individual interviews and questionnaire surveys will be conducted with several key freight movement stakeholders in Tennessee. The freight stakeholders and partners to be interviewed may include metropolitan planning organizations (MPOs), trucking companies, Chambers of commerce, trade organizations, Appalachian Regional Commission, Tennessee Valley Authority, etc. Other interviews will be aimed at truck drivers and passenger car drivers who are using the identified corridors on a regular basis. The stakeholders will initially be contacted prior to the field review schedules for this study. This will be done to identify issues that could be addressed by the multiple specialists included in the field review. Stakeholders will be contacted again following the identification of issues and potential solutions identified.

Task 6: Freight Traffic Operations Analysis

This task will focus on analyzing freight bottlenecks and optimizing traffic flow along Tennessee's critical freight corridors. Through employing microsimulation models and AI-driven traffic analysis, the project will assess the efficiency of current freight routes and identify key choke points. Real-time traffic data, coupled with historical trends, will be used to simulate various traffic scenarios and develop optimized routing strategies. The expected outcome is a significant reduction in delays and improved flow efficiency, particularly along I-40, I-24, I-75, I-65, I-55, I-240 and others.

Task 7: Statistical application, ML, and AI for Freight Safety and Efficiency

This task focuses on using AI and statistical models to enhance the safety and efficiency of freight movements across Tennessee. Crash prediction models will be developed to assess risk factors along critical freight routes, providing actionable insights to fleet managers for proactive risk mitigation. Additionally, AI-driven tools will be implemented to optimize last-mile delivery logistics, reducing delivery times and enhancing the safety of urban freight distribution. Perform Risk Management evaluation of freight corridors in Tennessee.

Task 8: AI, Machine Learning, and Predictive Analytics

This task aims to develop advanced AI models capable of predicting traffic patterns and optimizing freight routing in real-time. The AI system will be trained on a diverse dataset, including historical traffic data, weather patterns, incident reports, and freight schedules provided by TDOT and logistics partners. Through reinforcement of learning algorithms, the project will enable dynamic and adaptive route planning, ensuring that freight vehicles can avoid congestion and delays effectively.

Task 9: Solving Economic and Supply Chain Impact Analysis

This task will evaluate the economic benefits of optimized freight logistics by conducting a comprehensive cost-benefit analysis and supply chain impact assessment. By analyzing transportation costs, delivery reliability, and operational efficiency, the project will quantify the potential cost savings and environmental benefits of AI-driven routing solutions. The findings will be used to develop a Tennessee Freight Optimization Playbook that outlines policy recommendations and strategies for statewide adoption.

Task 10: Using AI and ML for Solving Freight Truck Parking

This task aims to utilize AI and machine learning to assess and optimize truck parking availability across Tennessee's Interstate highways. Leveraging GIS and remote sensing technologies, the project will identify underutilized parking areas and evaluate the feasibility of using interstate ramps for overflow parking. The project will develop a predictive model that forecasts parking demand at key locations based on real-time traffic data, historical parking trends, and freight schedules. The model will integrate with a real-time parking availability system to guide truck drivers to the nearest available spots, reducing unsafe parking practices and improving compliance with federal Hours-of-Service (HOS) regulations. The expected outcome is a reduction in illegal ramp parking incidents, improved safety and efficiency of freight movements, and actionable insights for future parking investments.

Task 11: Emissions Evaluation

Use EPA mobile source emissions model, MOVES to estimate truck emissions along the freight corridors and segments on a second-by-second basis in combination with VISSIM simulation software. The MOVES model estimations will be compared/combine with estimates from VISSIM microscopic traffic simulation model to obtain accurate emission results. The VISSIM/MOVES model will be used to determine freight transportation emissions factors for VOC, CO, NOx, SOx, and CO2.

Task 12: Develop Optimization and Analysis TOOLS for Freight Logistics Efficiency in Tennessee. Using findings from Tasks 1 to 11, the study will develop TOOLS for Optimization and Analysis of Freight Logistics Efficiency in Tennessee.